

Nonbacktracking Walks Settle the Degree 3,4,5 Long-Range Expansion Problem

Run `fa_banach_001`, agent `agent_lane_03`

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Status

Status: candidate full solution, likely valid, human review needed. This packet gives an affirmative answer to Problem 6.3 of Altschuler–Dodos–Tikhomirov–Tyros [1]: random d -regular graphs with $d \in \{3, 4, 5\}$ satisfy part (B) of the long-range expansion property with high probability.

Source Problem

The source paper defines a long-range expansion property $\text{Expan}(\alpha, \varepsilon, L)$ for d -regular graphs. Proposition 1.9 proves that random d -regular graphs satisfy part (A) for every $d \geq 3$, and part (B) for $d \geq 6$. The open problem is the missing low-degree range:

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6.3. Graphs with the long-range expansion property and degree $d \in \{3, 4, 5\}$. Proposition 1.9 shows that for any integer $d \geq 6$, a uniformly random d -regular graph satisfies part (B) of the long-range expansion property with high probability. A natural problem is to determine if this also holds true for regular graphs with degree $d \in \{3, 4, 5\}$. (Recall that part (A) is satisfied for any integer $d \geq 3$.)

Problem 6.3. *Is it true that for any integer $d \in \{3, 4, 5\}$, a uniformly random d -regular graph satisfies part (B) of the long-range expansion property with high probability?*

6.4. Explicit constructions of regular graphs with the long-range expansion property.

We use the source parameters

$$\alpha(d) = d^{-10^{11} \log d}, \quad \varepsilon = 0.2, \quad L(d) = 24\alpha(d)^{-1}.$$

Proof Intuition

The source proof of part (B) assumes a failure of long-range expansion and builds a mean-zero vector whose adjacency-walk averages are too large. The upper bound for ordinary adjacency walks is governed by roughly $(2\sqrt{d-1})^{2\ell} = 4^\ell(d-1)^\ell$. This is why the source argument only closes when $d \geq 6$.

For $d = 3, 4, 5$, count nonbacktracking walks instead. Nonbacktracking walk matrices are explicit polynomials in the adjacency matrix. Under Friedman’s spectral bound, their nontrivial growth is

essentially $(d-1)^{\ell/2}$, so the squared upper bound has base $d-1$. A failure of part (B) still forces the same lower base

$$\beta_d = \frac{(d-1-\varepsilon)^3}{d-1}.$$

For $d = 3, 4, 5$ and $\varepsilon = 0.2$, one has $\beta_d > d-1$. This strict exponential gap gives the contradiction.

Main Result

Theorem 1. *Let $d \in \{3, 4, 5\}$ and let*

$$\alpha = d^{-10^{11} \log d}, \quad \varepsilon = 0.2, \quad L = 24\alpha^{-1}.$$

There exists $\eta_d > 0$ such that every d -regular graph G satisfying

$$\lambda(G) \leq 2\sqrt{d-1} + \eta_d$$

satisfies part (B) of $\text{Expan}(\alpha, \varepsilon, L)$. Consequently, a uniformly random d -regular graph satisfies part (B) of this long-range expansion property with high probability.

Proof

Write $D = d-1$ and

$$\beta = \frac{(D-\varepsilon)^3}{D}.$$

For $d = 3, 4, 5$ and $\varepsilon = 0.2$, direct arithmetic gives

$$\frac{(D-\varepsilon)^3}{D^2} > 1.$$

Choose τ such that $\sqrt{D} < \tau < \sqrt{\beta}$, and set

$$\eta_d = \tau + \frac{D}{\tau} - 2\sqrt{D} > 0.$$

Lemma 1 (nonbacktracking spectral bound). *Let G be a d -regular graph with $\lambda(G) \leq 2\sqrt{D} + \eta_d$. For $k \geq 1$, let N_k be the matrix whose (u, v) entry is the number of nonbacktracking walks of length k from u to v . There is a constant $C_d < \infty$ such that, for every mean-zero $x \in \mathbb{R}^{V_G}$ and every $\ell \geq 1$,*

$$\left\| \sum_{k=1}^{\ell} N_k x \right\|_2^2 \leq C_d \ell^2 \tau^{2\ell} \|x\|_2^2.$$

Proof. The matrices N_k are polynomials $Q_k(A_G)$ in the adjacency matrix. They satisfy

$$Q_1(t) = t, \quad Q_2(t) = t^2 - d, \quad Q_k(t) = tQ_{k-1}(t) - DQ_{k-2}(t) \quad (k \geq 3).$$

Equivalently,

$$Q_k(t) = D^{k/2} U_k\left(\frac{t}{2\sqrt{D}}\right) - D^{(k-2)/2} U_{k-2}\left(\frac{t}{2\sqrt{D}}\right),$$

where U_k is the Chebyshev polynomial of the second kind. If $|t| \leq 2\sqrt{D}$, then $|U_k(t/(2\sqrt{D}))| \leq k+1$, so $|Q_k(t)| \leq 2(k+1)D^{k/2} \leq 2(k+1)\tau^k$.

If $2\sqrt{D} \leq |t| \leq 2\sqrt{D} + \eta_d = \tau + D/\tau$, write $|t| = r + D/r$ with $\sqrt{D} \leq r \leq \tau$. The same Chebyshev formula gives $|Q_k(t)| \leq 2(k+1)r^k \leq 2(k+1)\tau^k$. Thus every nontrivial eigenvalue of N_k has modulus at most $2(k+1)\tau^k$. Summing over $k \leq \ell$ and using the spectral theorem gives the asserted bound, after absorbing the elementary factor into $C_d\ell^2$. $\square$

We now prove the theorem. Suppose that part (B) fails for some nonempty $S \subseteq V_G$ and integer $\ell \geq 1$ satisfying $\alpha D^{\ell-1}|S| \leq 3|V_G|/4$. Let T be the set of edges from the source definition of part (B), and let R be the set of all endpoints of edges in T .

The double-counting estimate from the source proof remains unchanged:

$$|T| \leq |S| \frac{d}{L(d-2)} \left(\frac{D}{D-\varepsilon} \right)^\ell.$$

Since $|T|/d \leq |R| \leq 2|T|$, the source argument also gives

$$|R| \leq \frac{|V_G|}{2}, \quad \frac{|R^c|}{|R|} \geq |V_G| \frac{(d-2)L}{4d|S|} \left(\frac{D-\varepsilon}{D} \right)^\ell.$$

Define the mean-zero unit vector

$$x = \sqrt{\frac{|R^c|}{|R|}} \frac{\mathbf{1}_R}{\sqrt{|V_G|}} - \sqrt{\frac{|R|}{|R^c|}} \frac{\mathbf{1}_{R^c}}{\sqrt{|V_G|}}.$$

Set $Z = \sum_{k=1}^\ell N_k x$.

For every $v \in S$, failure of part (B) says that more than $L(D-\varepsilon)^\ell$ edges of T are within distance $\ell-1$ of v . Passing from edges to endpoints loses at most a factor d . If the endpoint is v itself, discard it; this loses at most one endpoint and is absorbed by the fixed source choice of L . Every remaining endpoint is reached by a shortest path of length between 1 and ℓ , hence by a nonbacktracking path. Thus, with slack,

$$\sum_{w \in R} \sum_{k=1}^\ell (N_k)_{vw} \geq \frac{L}{2d} (D-\varepsilon)^\ell.$$

On the other hand, the total number of nonbacktracking walks of lengths $1, \dots, \ell$ from a fixed vertex is at most

$$d \sum_{k=1}^\ell D^{k-1} \leq \frac{d}{d-2} D^\ell.$$

Combining these inequalities with the lower bound for $|R^c|/|R|$ gives, for all $v \in S$,

$$Z_v \geq \frac{L^{3/2}\sqrt{d-2}}{8d^{3/2}\sqrt{|S|}} \left(\frac{(D-\varepsilon)^3}{D} \right)^{\ell/2}.$$

Here the factor $1/8$ records both the harmless endpoint discard and the subtraction of the R^c contribution; with the source values of α, ε, L , the needed domination is a fixed arithmetic inequality for $d = 3, 4, 5$.

Consequently,

$$\|Z\|_2^2 \geq \frac{L^3(d-2)}{64d^3} \beta^\ell.$$

The nonbacktracking spectral lemma gives the opposite estimate

$$\|Z\|_2^2 \leq C_d \ell^2 \tau^{2\ell}.$$

Since $\tau^2 < \beta$, the right-hand side is exponentially smaller than the left-hand side as ℓ grows, and the remaining bounded range of ℓ is covered by the fixed source choice $L = 24\alpha^{-1}$. The script `code/verify_constants.py` checks these finite degree-3, 4, 5 inequalities using the explicit choices

$$\tau = \frac{\sqrt{D} + \sqrt{\beta}}{2}.$$

This contradiction proves the deterministic statement.

Finally, Friedman’s theorem says that for every fixed $d \geq 3$ and every $\eta > 0$, a uniformly random d -regular graph satisfies $\lambda(G) \leq 2\sqrt{d-1} + \eta$ with high probability. Applying this with $\eta = \eta_d$ completes the proof of Problem 6.3.

Verification Notes

The verifier prints

$$\beta_3 = 2.916, \quad \beta_4 \approx 7.3173, \quad \beta_5 \approx 13.718,$$

which are strictly larger than $d-1 = 2, 3, 4$, respectively. It also checks, in logarithmic arithmetic, that the negative term in the lower bound is absorbed and that the lower/upper spectral estimates cannot be compatible for any $\ell \geq 2$.

Novelty and Duplicate Check

The run indexes were searched on 2026-07-03 for arXiv:2410.04394, the paper title, long-range expansion, nonlinear spectral gaps, Problem 6.3, random regular graph degree 3, 4, 5, and nonbacktracking variants. No prior packet for this target was found. External web searches on the same date for the exact title, arXiv id, long-range expansion degree 3, 4, 5, and Problem 3 phrases surfaced the original arXiv page but no later answer. The proof uses only standard nonbacktracking walk polynomials and Friedman’s theorem.

Human Review Recommendation

Review the nonbacktracking spectral lemma and the lower-bound transfer from the source adjacency-walk proof. If those are accepted, the proof gives a complete affirmative answer to Problem 6.3.

References

- [1] D. J. Altschuler, P. Dodos, K. Tikhomirov, and K. Tyros, *A combinatorial approach to nonlinear spectral gaps*, arXiv:2410.04394, 2024.
- [2] J. Friedman, *A proof of Alon’s second eigenvalue conjecture and related problems*, *Memoirs of the American Mathematical Society* 195 (2008), no. 910.